State space system

To capture this model in a state space system, a system of nonlinear differential equations is made. Because there are different modes in the system, discrete variables are used to turn the modes on and of. The terms that are changing in that case are denoted by Δ.

$$\begin{pmatrix} \dot{T}_r \\ \dot{T}_2 \\ \dot{T}_B \\ \dot{\theta} \\ \ddot{\theta} \\ \dot{h} \end{pmatrix} = \begin{pmatrix} \left(\frac{1}{R_2}(T_2 - T_r) + \frac{1}{R(\theta)}(T_0 - T_r)\right)/C_r \\ \left(\frac{1}{R_2}(T_r - T_2) - T_2 f_1\right)/C_2 \\ \left(I(t)A\eta\frac{R_B}{R_B+R_1}\right)/C_B \\ 0 \\ 0 \\ -\frac{1}{A_T}\left(\frac{1}{R_T}h\rho g\right) \end{pmatrix} + \Delta \tag{1}$$

Heating

When the heating mode is turned on, the Δ terms that are added are:

$$\Delta = \begin{pmatrix} \dot{T}_r \\ \dot{T}_2 \\ \dot{T}_B \\ \dot{\theta} \\ \ddot{\theta} \\ \dot{h} \end{pmatrix} = \begin{pmatrix} 0 \\ (T_2 - T_B)\, f_1/C_B \\ (T_B f_1)/C_2 \\ 0 \\ 0 \\ 0 \end{pmatrix} \tag{2}$$

Cooling

When the cooling mode is turned on, the Δ terms that are added are:

$$\Delta = \begin{pmatrix} \dot{T}_r \\ \dot{T}_2 \\ \dot{T}_B \\ \dot{\theta} \\ \ddot{\theta} \\ \dot{h} \end{pmatrix} = \begin{pmatrix} 0 \\ T_2 f_1/C_2 \\ 0 \\ 0 \\ 0 \\ \frac{1}{A_T}\omega_1 \end{pmatrix} \tag{3}$$

Window opening

When the window is opening, the Δ terms that are added are:

$$\Delta = \begin{pmatrix} \dot{T}_r \\ \dot{T}_2 \\ \dot{T}_B \\ \dot{\theta} \\ \ddot{\theta} \\ \dot{h} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{4\omega_0}{A_c \cos\frac{\pi}{4} + \frac{\theta}{2} l} \\ 0 \\ 0 \end{pmatrix} \tag{4}$$

Window closing

When the window is closing, the Δ terms that are added are:

$$\Delta = \begin{pmatrix} \dot{T}_r \\ \dot{T}_2 \\ \dot{T}_B \\ \dot{\theta} \\ \ddot{\theta} \\ \dot{h} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \ddot{\theta} dt \\ \Big(-\frac{l}{2} mg \sin\theta \\ \quad + \Big(A_c^2 R_0 \frac{1}{4} l \cos(\frac{\pi}{4} + \frac{\theta}{2})\dot{\theta} \\ \quad + A_c P_0\Big) \sin(\frac{\pi}{4} - \frac{\theta}{2})\Big)/I \\ 0 \end{pmatrix} \tag{5}$$

Moore machine controller - extended

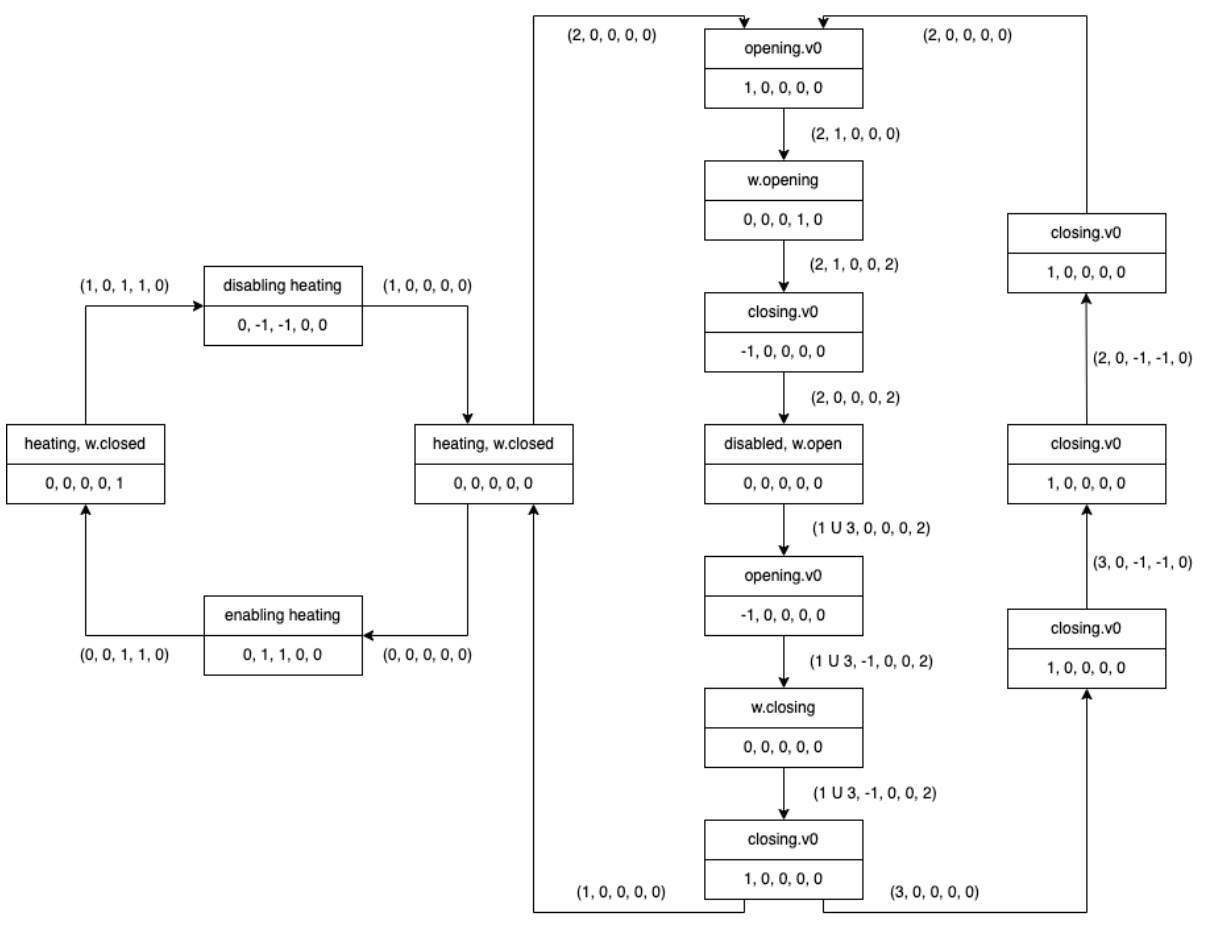

Figure 1: Moore machine of extended controller

Moore machine controller - reduced

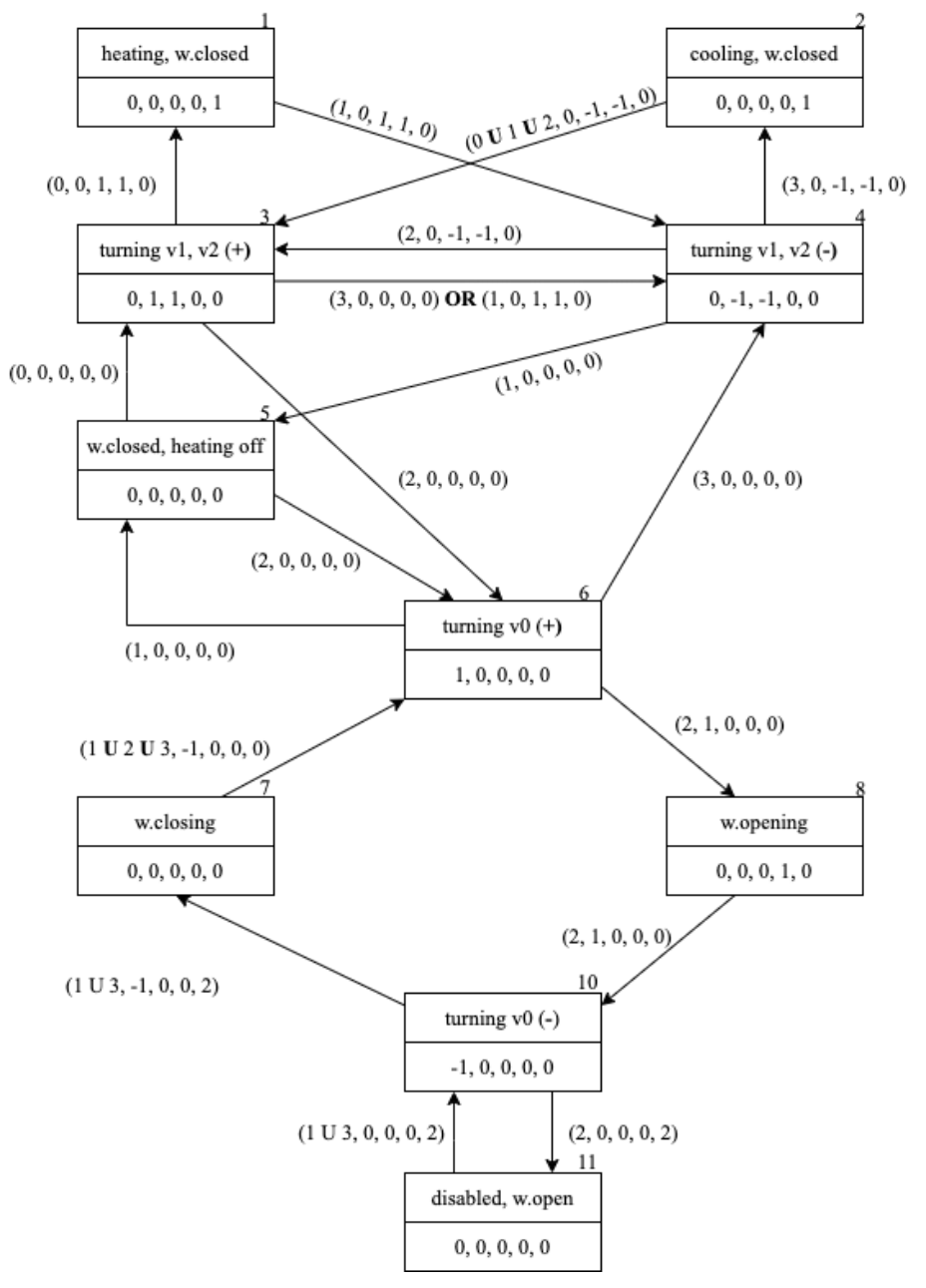

Figure 2: Moore machine of controller with reduced amount of nodes

Hybrid System

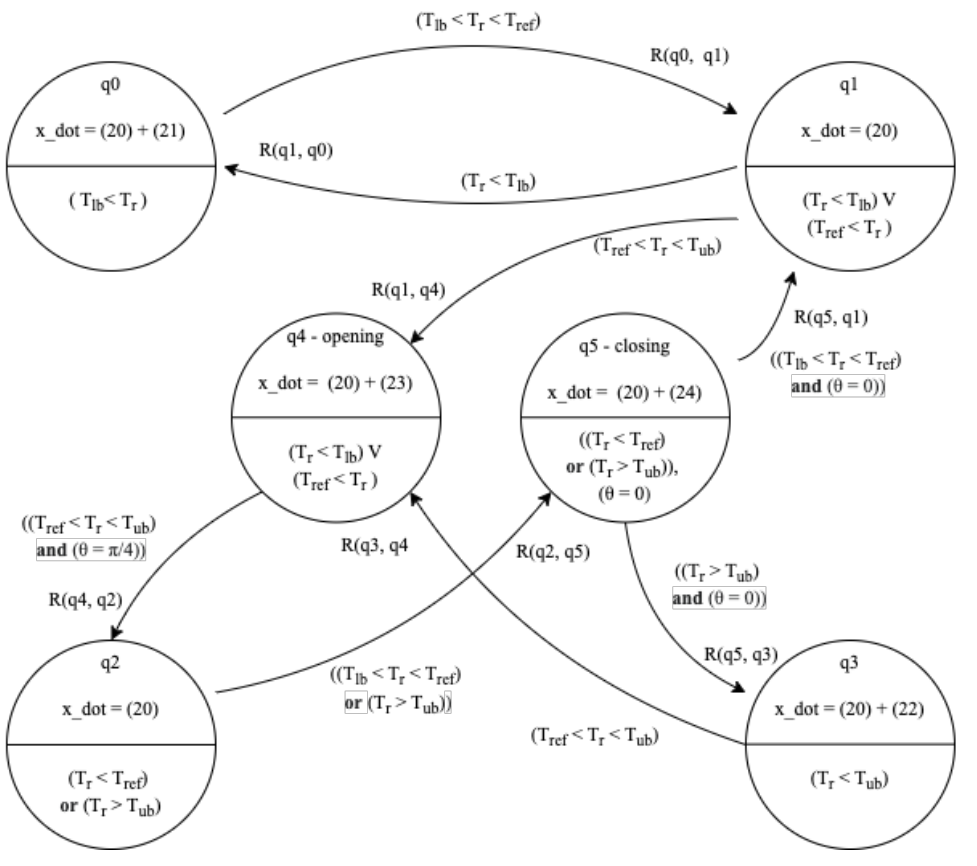

Figure 3: The Hybrid system diagram

Simulation Results

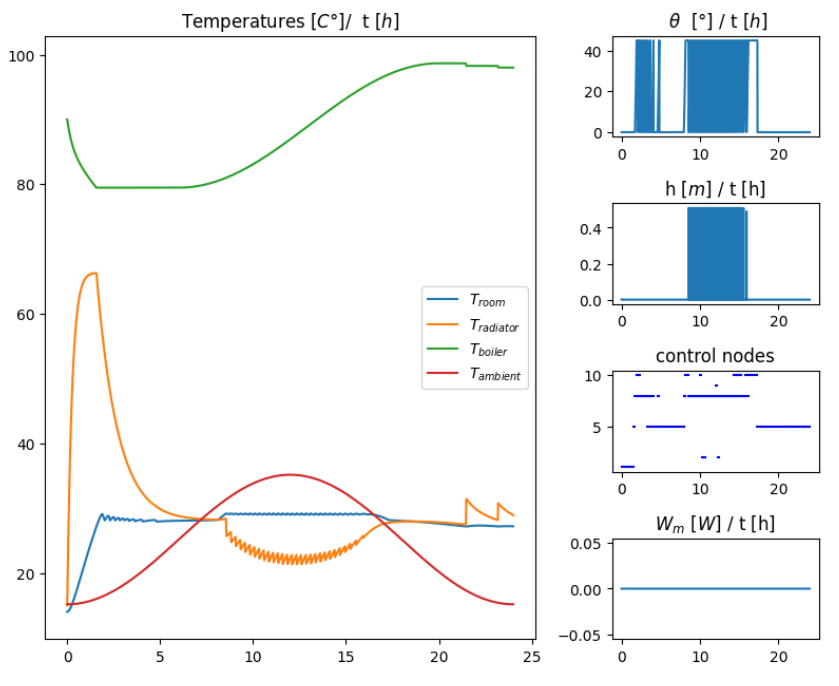

Figure 4: Simulation with a varying T_0